

Evaluating brain spectral and connectivity differences between silent mind-wandering and trance states

Helané Wahbeh^{a,*}, Cedric Cannard^a, Michael Kriegsman^a, and Arnaud Delorme^{a,b}

^aResearch Department, Institute of Noetic Sciences, Petaluma, CA, United States

^bUniversity of California, San Diego, CA, United States

*Corresponding author: e-mail address: hwahbeh@noetic.org

Abstract

Trance is an altered state of consciousness characterized by alterations in cognition. In general, trance states induce mental silence (i.e., cognitive thought reduction), and mental silence can induce trance states. Conversely, mind-wandering is the mind's propensity to stray its attention away from the task at hand and toward content irrelevant to the current moment, and its main component is inner speech. Building on the previous literature on mental silence and trance states and incorporating inverse source reconstruction advances, the study's objectives were to evaluate differences between trance and mind-wandering states using: (1) electroencephalography (EEG) power spectra at the electrode level, (2) power spectra at the area level (source reconstructed signal), and (3) EEG functional connectivity between these areas (i.e., how they interact). The relationship between subjective trance depths ratings and whole-brain connectivity during trance was also evaluated. Spectral analyses revealed increased delta and theta power in the frontal region and increased gamma in the centro-parietal region during mind-wandering, whereas trance showed increased beta and gamma power in the frontal region. Power spectra at the area level and pairwise comparisons of the connectivity between these areas demonstrated no significant difference between the two states. However, subjective trance depth ratings were inversely correlated with whole-brain connectivity in all frequency bands (i.e., deeper trance is associated with less large-scale connectivity). Trance allows one to enter mentally silent states and explore their neurophenomenological processes. Limitations and future directions are discussed.

Keywords

Silence, Trance, Mind-wandering, EEG, Connectivity

1 Introduction

1.1 The sound of silence

1.1.1 External sound

The faculty of hearing brings us exquisite soothing sounds such as waterfalls, babbling brooks, and laughter, and also stress-inducing sounds like jackhammers, city traffic, and car horns. External sounds can influence our mood and health, as observed in multiple studies on music ([Fancourt et al., 2014](#)) and noise pollution ([Domingo-Pueyo et al., 2016](#)). Rarely do humans experience complete silence in their environment. When these louder stimuli are removed, our hearing can become so sensitive that we can detect low-decibel (dB) sounds ([Cox, 2014](#)). Extremely quiet places include Microsoft's Redmond sound-dampening chamber at -20 dB, the Guinness Book of Records' quietest spot in the world ([Lilly, 2019](#)), and Orfield Laboratories at -9.4 dB ([Chhaya, 2013](#)). For comparison, a quiet bedroom may measure around 30 dB. Zero dB corresponds to a sound pressure of 2×10^{-5} Pascals and the standard threshold for human hearing ([Nave, 2016](#)). These sound-dampening or anechoic chambers are so quiet that people can hear their hearts beating, blood moving in their veins, or their stomachs digesting food ([Puiu, 2021](#)).

1.1.2 External silence

External silence also impacts us. For example, an absence of sound experienced in anechoic chambers leads to an altered state where people without any psychopathology experience auditory hallucinations ([Mason and Brady, 2009](#)). Sensory deprivation in float tanks, in which individuals float in salinated water at body temperature without visual or auditory stimulation, creates deep relaxation and trance states ([Walters, 2020](#)). In the OVO Whole-Body Perceptual Deprivation chamber, participants experienced time perception changes ([Glicksohn et al., 2017](#)) and increased absorption, which is the person's disposition to become absorbed in their mental imagery and is characteristic of trance states ([Ben-Soussan et al., 2019](#)).

1.1.3 Internal sound

Interestingly, we also experience another type of "sound." That is the sound of our thoughts or inner speech that we can think of as internal mental sounds. Our thoughts can be loud when we pay attention to them (e.g., during meditative states). Most have experienced the sound of thoughts as mind-wandering, the mind's propensity to stray its attention away from the task at hand and toward content irrelevant to the current moment ([Alderson-Day and Fernyhough, 2015](#)). The central dimension of mind-wandering is inner speech, that is, verbal thoughts as opposed to visual or nonverbal mental processes. A large part of life is spent mind-wandering, suggesting it plays an essential role in brain function and cognition ([Travis, 2021](#)).

In the context of sustained attention and aptitude tests (e.g., working memory, reading), mind-wandering has been consistently associated with reduced performance and, therefore, considered a “failure” of cognitive control. Similarly, mind-wandering is associated with poor mental health, expressed as mental rumination about past negative memories and worry about the future (Vago and Zeidan, 2016). One study evaluating over 5,000 people from 83 countries in their daily lives found that mind-wandering correlated with decreased happiness compared to participants that were not mind-wandering when surveyed (Killingsworth and Gilbert, 2010).

However, other studies share that mind-wandering is not intrinsically detrimental but that its negative impact is a consequence of the mind-wandering content (positive or negative valence). Negative mind-wandering may result from prior negative mood and mental health but also cause it. In one study, researchers found that when participants felt sadness, mind-wandering worsened their mood, and their mind-wandering content was negative (Poerio et al., 2013). People with healthy and satisfying lives have predominantly positive content when their minds wander and may use it as a strategy to regulate their emotions during negative situations (e.g., positive daydreaming to escape the situation), make better predictions by simulating alternative situations through imagination, or nurture creativity (Baird et al., 2012; Franklin et al., 2013; Mooneyham and Schooler, 2013; Travis, 2021).

1.1.4 Internal silence

Mental silence, or the absence of cognitive thought, is experienced through many methods. One common way to experience mental silence is through meditation practice. Mental silence (and mind-wandering) is observed during meditation because most meditation practices include bringing awareness to mental activity and positive and negative patterns that arise spontaneously. It provides “a unique and intimate opportunity to closely examine the nature of the wandering mind by cultivating an awareness of ongoing thought patterns while simultaneously aiming to cultivate equanimity (evenness of temper or disposition) and compassion toward the content of thoughts, interpretations, and bodily sensations” (Brandmeyer and Delorme, 2021, p. 36). In general, meditation accompanied by increased mental silence and focused attention impart the greatest benefit (Carmody and Baer, 2008; Delorme and Brandmeyer, 2019; Jacobs et al., 2011; Mrazek et al., 2013; Sahdra et al., 2011).

Mental silence is the desired experience of many spiritual practices that incorporate meditation. Meditators typically aim to let go of attention to their thoughts and focus them on an object of interest (e.g., mantra, body scanning, breath) or simply observe the next thought without judgment or attachment to that thought. As practitioners develop, their sense of self shifts from thoughts anchored on the body to a sense of beingness that includes but also transcends the physical body (Cahn and Polich, 2006). The transcendent state where the person experiences profound unification is characterized by thoughtless inner awareness and often accompanied by feelings of bliss and joy. Interestingly, multiple traditions (e.g., Buddhist, Vedic, Christian, and secular meditation) describe this state very similarly (Wahbeh et al., 2018b). For example, the Vedic tradition describes this transcendent state as proceeding from gross to subtle mental activity and then to bliss or reaching

the silent state of awareness where there is no phenomenological content and a loss of any sense of the individual self or individuality. The point of transcendence through meditative practice is to experience this contentless awareness, silent and awake within itself (Wahbeh et al., 2018b, p. 21).

Few studies evaluate the health benefits of mental silence because rigorously assessing mental silence requires participants to be highly experienced and able to detect or induce internal silence. Researchers have evaluated mental silence via Sahaja yoga practice that purposefully engages in mental silence and found benefits for menopausal hot flashes (Manocha et al., 2007), work-related stress and depression symptoms (Manocha et al., 2011), anxiety and well-being (Hendriks, 2018), and increased gray matter density and coherence changes in the right inferior frontal gyrus which correlated with subjective well-being scores (Dodich et al., 2019). People in trance states can also experience mental silence. For example, psychography practitioners produce written material while in a trance state (Rocha et al., 2014). They report deep relaxation without mental distractions (e.g., mind-wandering) compared to the control condition and having no awareness of what their hand is writing, phenomenological differences supported by the study's functional magnetic resonance imaging (fMRI) data (Peres et al., 2012).

Sound and silence, both external and internal, play a role in health and well-being. This paper explores mental silence (or the absence of internal mental sounds) during a trance state, in contrast to mind-wandering (or internal noise).

1.2 Trance and its relationship to mental silence and mind-wandering

The American Psychological Association defines altered states of consciousness as a highly subjective “state of psychological functioning that is significantly different from that experienced in ordinary states of consciousness...[and] tends to be characterized by altered levels of self-awareness, affect, reality testing, orientation to time and place, wakefulness, responsiveness to external stimuli, or memorability, or by a sense of ecstasy, boundlessness, or unity with the universe” (American Psychological Association, 2022). Numerous situations, such as drug use, hypnosis, meditation, and sensory deprivation, can trigger altered states of consciousness. Some argue that the term “altered states of consciousness” is erroneous because it focuses on the content of our experience and thus should actually be called altered states of phenomenology, representing a shift from our normal daily functioning and subjective experience (Rock and Krippner, 2012).

Trance is an altered state of consciousness characterized by alterations in cognition. While trance experiences are highly subjective, trance states share common phenomena, such as loss of the sense of self, time, or the body, loss of control, changes in emotional expression, ineffability, perceptual distortion, diminished perception of the external surroundings, or heightened focus on a particular aspect of the external environment, and hyper-suggestibility in the context of hypnosis (Cardena et al., 2009; Pekala et al., 2010; Winkelman, 1986). For example, someone in a trance

may not feel the passage of time, may feel that the boundaries of their body have dissolved, a slowness or cessation of thought, and vivid visual imagery (Flor-Henry et al., 2017). In general, trance states induce mental silence (i.e., cognitive thought reduction through decreased consciousness; see Wahbeh et al. (2019) for a discussion of levels of consciousness during trance states), and mental silence can induce trance states (e.g., sensory deprivation; Walters, 2020).

Trance induction occurs in various ways. One way is through auditory or visual “driving,” where rhythmic auditory or visual stimuli, such as chanting, music, drumming, binaural beats, or pulsed photic stimulation, entrain the brain to slow frequency oscillations or states (Winkelman, 1986). Another trance induction technique is fasting, which ancestral, traditional, and contemporary indigenous people use in spiritual ceremonies (Thomason, 2010; Winkelman, 2011). In spiritual traditions such as mediumship and trance channeling, individuals can enter into various depths of trance states at will (Castillo, 2003; Seligman, 2005; Seligman and Kirmayer, 2008; Wahbeh et al., 2018a; Wahbeh and Butzer, 2020). Trance channeling can be described as the belief that “the channeler goes into a trance state at will (the depth of the trance may vary), and the disincarnate entity/spirit uses the channeler’s body with permission to communicate *directly* through the channeler’s voice, body movements, etc. (rather than the channeler receiving information mentally or otherwise and then relaying what is being received)” (Wahbeh et al., 2019, p. 5). Mediums believe they communicate with non-physical beings, typically deceased humans. Regardless of these claims’ veracity, these practitioners enter into various trance levels at will (Wahbeh et al., 2018, 2019; Wahbeh and Butzer, 2020). Hypnosis is a common way of inducing trance, and the hyper-suggestibility experienced during trance states can support healing (Pekala et al., 2010; Walters, 2020). For example, in the hyper-suggestible hypnotic trance state, patients are given suggestions, like refraining from cigarette smoking (Tahiri et al., 2012) or reducing their pain (Thompson et al., 2019). Other situations and states where trance is observed include meditation (Castillo, 2003), sleep (during hypnagogic or hypnopompic periods prior to sleep and waking, respectively), sexual restriction, extensive motor behavior (e.g., marathon running or vigorous dancing), endogenous opiate release, hallucinogen intake, alcohol use, and temporal lobe syndromes (e.g., seizures, tumors, infarction; Winkelman, 1986).

These inductive methods differ in their approach to achieving the trance state. For example, Mooneyham & Schooler showed that hypnosis and meditation were similar in terms of relaxation and absorption (Mooneyham and Schooler, 2015). In contrast, they differ in their respective direction of attention, outward for hypnosis (e.g., the hypnotist’s voice or watching a pendulum) and inward for meditation.

In addition, trance states appear graded from light depth, such as when driving in automatic pilot or daytime reverie (i.e., daydreaming), to a deeper depth, such as during hypnosis (Pekala et al., 2010) or trance channeling (Wahbeh et al., 2019). Buhrman proposes that different absorption stages reflect variations in consciousness during trance accompanied by varying levels of amnesia after the trance period (Buhrman, 1997). People who voluntarily enter trance states, such as trance channelers, report a spectrum of trance depth or awareness during their channeling

experiences. For example, one study reported a spectrum of conscious awareness among channelers who were completely unconscious or semiconscious and a theme of amnesia to what occurred during the trance state (Hughes, 1991). Another team of researchers in Brazil found similar variations in trance depth in “Spiritist mediums” (Negro et al., 2002). We also asked two groups of trance channelers to rate their awareness level on a scale from 0 to 100, with zero representing being fully conscious and aware and 100 representing being completely unconscious and not aware, and found an average score of 47.7 ± 33.6 (Wahbeh et al., 2019; Wahbeh and Butzer, 2020). Thus, on average, apparent trance channelers did not report being wholly unconscious during their trance but in a moderate trance with marked variation reflected by the large standard deviation of the scores. Notably, the depth of trance is not correlated with the effectiveness of modalities that use trance for therapeutic purposes (Pekala et al., 2010). Thus, while trance depth appears graded subjectively, trance effects are likely not dependent on the depth (i.e., someone may subjectively report a light trance but obtain mental silence and/or changes in behavior from suggestions provided during hypnotic trance).

Fig. 1 graphically represents the relationships between trance, external and internal sounds, and external and internal silence.

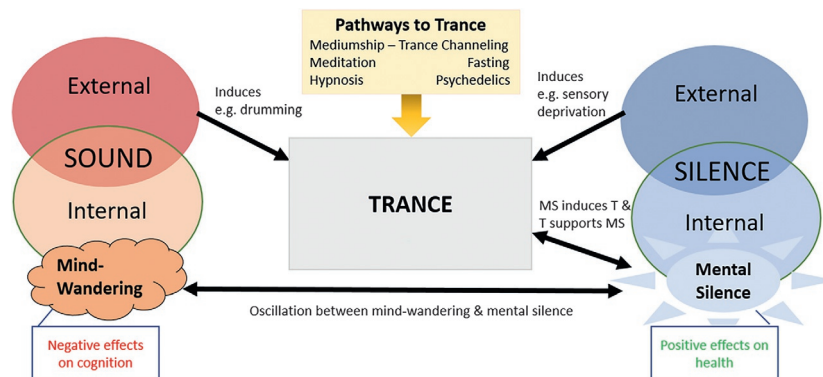


FIG. 1

Relationships between mental silence, mind-wandering, and trance. The human mind can oscillate between mind-wandering and mental silence. Trance is induced by external sound and silence and internal silence. Trance, in turn, supports internal/mental silence. Many methods, including mediumship, trance channeling, and meditation, induce trance states. Meditation studies help us understand the relationship between trance, mind-wandering, and mental silence because of the numerous studies on it and meditation’s inherent practice of observing mental activity. In general, negative mind-wandering is a detriment to cognition and health, whereas mental silence imparts mental and physical benefits. MS, mental silence; T, trance.

1.3 Neural correlates of silence

While there may be physical and mental health benefits to mental silence, one question concerning mental silence and trance is whether we can identify physiological correlates of mental silence, specifically in the brain.

1.3.1 *Neuroimaging of external noise*

The brain's response to auditory stimuli and external silence is well-documented in electroencephalography (EEG) research (Picton, 2010; Savers et al., 1974). For example, participants completing problem-solving tasks in the laboratory during three varying noise conditions (no noise, noise cancelation within a noisy environment, and noise) demonstrate significantly different EEG patterns (Pieper et al., 2021). Responses to auditory stimuli are also used to assess consciousness levels in patients under anesthesia during medical procedures (Plourde, 2006). Thanks to brain-computer interface technologies, researchers have been able to classify EEG event-related potential responses to pleasant, neutral, and unpleasant auditory stimuli as well (Hettich et al., 2016). Researchers have also extensively studied speech perception and processing. Recent groundbreaking studies demonstrate that the brain sequences the three most recently heard sounds in parallel (both the order and the content of the phonemes) and maintains it long after the sensory input is dissipated (Gwilliams et al., 2022).

While evaluating how the brain processes auditory stimuli (external noise) compared to silence is common, little is known about brain processes and dynamics during external silence. To our knowledge, the only line of work in this area concerns the investigation of inner speech (also called covert speech or verbal thinking), a form of mind-wandering. While no consensus exists regarding inner speech mechanisms, this research field has generally found that it plays an essential role in cognitive development, executive functions, and psychopathology (Alderson-Day and Fernyhough, 2015). Inner speech is now thought to have an abstract and concrete (embodied) dimension, and it should be interpreted as an action-perception (sensorimotor) phenomenon as opposed to a predictive control phenomenon as previously thought. Inner speech is similar to overt language (i.e., language, sensorimotor and cognitive control networks), involving operations in multiple modalities, such as phonation, articulation, and gesture associated with language, but with an additional recruitment of inhibitory circuits to keep the speech covert as opposed to overt. A detailed description of the intersection of EEG and external noise and silence is beyond the scope of this work but warrants mention.

1.3.2 *Neuroimaging and internal silence*

1.3.2.1 Functional neuroimaging

The neural correlates of internal silence are even less understood as far as we know, except for a few studies in the meditation research context. For example, fMRI findings suggest that meditative states (e.g., during Sahaja yoga) are associated with decreased bilateral activity in the deep frontal and temporal regions. This activation

changes during the deepest meditation states to the right inferior frontal cortex/right insula and right middle/superior temporal cortex becoming more activated (Hernández et al., 2015). A later study by the same researchers found that gray matter volume in the medial prefrontal cortex, crucial for top-down cognitive, emotional, and attention control, was positively correlated with subjective mental silence ratings (Hernández et al., 2018).

1.3.2.2 EEG

Unlike the expensive, bulky, and time-consuming fMRI measure, the high temporal resolution, non-invasiveness, portability, and relatively low cost have made EEG a valuable scientific tool for studying psychological states and examining the underlying brain correlates, dynamics, and mechanisms. Researchers have evaluated EEG data during mental silence in multiple meditation studies. Thoughtless transcendent states were evaluated in a systematic review of 25 studies where transcendent states are uniformly described as a “silent state of pure consciousness where there is no phenomenological content and a loss of any sense of individual self or duality” despite the varied meditation tradition (e.g., Buddhist, Vedic—Transcendental Meditation and Yoga, Christian; Wahbeh et al., 2018b, p. 20). The review showed that mental silence state was positively associated with alpha power and EEG functional connectivity (phase coherence, i.e., greater alpha power and phase coherence when there is greater mental silence, Wahbeh et al., 2018b).

One included study observed gamma power decreases in centro-parietal regions during thoughtless emptiness (mental silence) relative to a focused attention meditation, decreased alpha and beta power in parietal areas relative to open monitoring meditation, and decreased delta and theta power relative to the closed eyes resting condition (i.e., mental noise; Hinterberger et al., 2014). These mental silence states have also been associated with gamma band EEG (Lutz et al., 2004), although these results are controversial considering the presence of muscle activity in the gamma band. More recently, Rodriguez-Larios et al. (2021) compared EEG profiles of experienced and novice meditators in relation to their subjective mind-wandering and attention-focus ratings. Experienced meditators had significant decreases in alpha and beta band power during mental silence compared to a mind-wandering control condition. The novice meditators could not achieve mental silence and experienced mind-wandering during meditation, which was related to a significant increase in alpha and beta power (Rodriguez-Larios et al., 2021).

1.3.3 EEG and mental silence during trance states

Winkelman proposed that the common element across trance induction techniques is that they induce a parasympathetic dominant state, leading to slow frequency patterns in the frontal cortex modulated by the subcortical structures of the limbic system (Winkelman, 1986). A systematic review summarized that trance states showed

significantly increased EEG spectral power in the theta (3–7 Hz), alpha (8–13 Hz), and beta (13–30 Hz) frequency bands compared to control conditions (Bastos et al., 2015). Two studies published since the review also observed increases in alpha power in one frontal electrode (F4) region immediately after trance compared to controls (Bastos et al., 2016) and averaged over 11 scalp sites in participants during trance compared to rest (linked Fp1–Fp2, F7, Fz, F8, C3, C4, T5, Pz, T6, O1, and O2; Kawai et al., 2017), and higher beta in F7 and F8 during trance relative to non-medium controls (Bastos et al., 2016).

Further, voluntarily quieting the mind facilitates the trance experience (Castillo, 1995). For example, experienced mediums closed their eyes and actively ignored external and internal stimuli to achieve a “state of emptiness,” suggesting that cognitive control is required for a quiet mind and supports mediumistic trance induction (Mainieri et al., 2017). In addition, Ben-Soussan et al. (2019) found during a sensory deprivation study that increased delta and theta power in the left inferior frontal cortex (with a peak in the sub-lobar of the left insula) were positively associated with trait absorption using exact-low resolution electromagnetic tomography (eLORETA; Ben-Soussan et al., 2019).

As an initial effort to evaluate EEG and trance, we conducted a cross-sectional study (Wahbeh et al., 2019) of 13 trance channelers alternating between a trance and mind-wandering state multiple times (six 5-min blocks per condition in total repeated on consecutive days). Our general linear modeling (GLM) analysis of the EEG power spectral density at the electrode level did not show significant differences between the trance and mind-wandering states. There was a trend toward significance at 13 Hz in uncorrected statistical maps, but it did not survive the spatiotemporal cluster correction for multiple comparisons using the state-of-the-art threshold-free cluster enhancement (TFCE) transformation (Wahbeh et al., 2019, p. 12). The post-hoc analysis of the interaction between collapsed EEG electrodes into broader regions (frontal, frontal right and left, central, central right and left, and parietal right and left) also did not result in any significant differences between trance and mind-wandering (Wahbeh et al., 2019, p. 13).

However, most spectral analyses of scalp EEG signals are hindered by the fact that cortex-generated electric fields are subject to volume conduction effects (i.e., distortion of the electric fields as they traverse brain tissues and the skull), leading to cross-talk or correlated noise in the scalp signal (Haufe and Ewald, 2019; Nunez and Srinivasan, 2006). Unfortunately, while scalp EEG is widely used, conduction effects can result in biased estimations of power spectral density, especially when looking at EEG phase coherence (functional connectivity) between electrodes separated by a short distance (Nunez and Srinivasan, 2006).

Volume conduction and other methodological issues, such as variations in data processing, statistical methods, brain anatomy, demographics, and interventions, often result in ambiguous results from systematic reviews and meta-analyses. For example, reviews of EEG meditation studies are inconclusive, but one that encompassed 64 studies from 1965 to 2005 reported a general increase in alpha power in

various meditation traditions (Cahn and Polich, 2006). However, it is uncertain if this increase is explicitly related to meditation, especially considering the heterogeneity of meditation practices, or perhaps to increased relaxation. Similarly, the absence of differences in the preliminary study (Wahbeh et al., 2019) may be due to the inability of standard spectral measures at the electrode level to adequately evaluate differences in functional connectivity dynamics (i.e., how different regions interact) between trance and mind-wandering. Furthermore, electrode-level EEG analyses (and especially on signals averaged to reference compared to reference-free Surface-Laplacian transformations; Nunez and Srinivasan, 2006) have poor spatial resolution and do not offer much information regarding local/global dynamics, neural networks, EEG sources, and the mechanisms underlying the state of interest.

1.4 EEG analysis advances support mental silence evaluation

Advances in signal processing techniques, such as EEG source separation, localization, and connectivity, provide other avenues to better understand mental silence and trance. Localizing the source of EEG fields (i.e., the neural population generating the signal inside the brain) requires solving the inverse problem. As of yet, no direct source localization solution exists. Thanks to recent computing advances, approximation techniques using prior source activity assumptions, such as spatial smoothness constraints (Pascual-Marqui et al., 1994), sparsity penalties (Ding and He, 2008; Haufe et al., 2008; Owen et al., 2012), or temporal assumptions by brain location (Gross et al., 2001), are now available in open-source platforms. These now allow for advanced analysis of spatiotemporal dynamics (Haufe et al., 2011).

EEG connectivity analysis is another technological advance that allows the estimation of interactions between different brain region sources and plays a crucial role in understanding general brain functions and spatiotemporal dynamics. Popular approaches estimate these interactions using the cross-spectrum (Nunez et al., 1997), Granger causality (Kamiński and Blinowska, 1991), dynamic causal modeling (Kiebel et al., 2008; Stephan et al., 2007), or phase/amplitude correlations of brain rhythms (Brookes et al., 2011). Popular MRI atlases typically define the brain regions of interest and source localization performed using the standardized/exact low-resolution brain electromagnetic tomography (sLoreta/eLoreta, respectively; Pascual-Marqui, 2002; Pascual-Marqui et al., 1999) or Beamforming (Jonmohamadi et al., 2014). Connectivity is then performed between all pairs of brain regions using Granger causality or direct transfer entropy, for example. These inverse methods allow the detection of false connectivity on both electrode and source space (Haufe et al., 2013).

However, source localization and reconstruction require validation (Haufe and Ewald, 2019). Progress on this front has been made with studies comparing EEG with fMRI data (Grova et al., 2008), phantom studies (Baillet et al., 2001), or simulation studies (Haufe, 2015; Huang et al., 2016), and also the standardized benchmark by

[Haufe and Ewald \(2019\)](#). Despite the lack of a definitive inverse problem solution, these methods facilitate assessing how these source localization and reconstruction techniques perform and increase our knowledge.

1.5 Evaluating brain connectivity during trance compared to mind-wandering

Building on the previous literature on mental silence and trance states and incorporating these inverse source reconstruction advances, the objectives of this secondary study were to evaluate differences between trance and mind-wandering states using: (1) power spectra at the electrode level (as previously analyzed but using novel automated signal processing tools); (2) power spectra at the area level, after source localization and reconstruction; and (3) EEG connectivity between these areas (how activity at these reconstructed areas interact). In addition, the relationship between subjective, self-reported trance depths ratings and whole-brain connectivity during the trance condition was evaluated.

We did not anticipate significant differences in the spectral analysis because we did not observe differences in the original study ([Wahbeh et al., 2019](#)). Based on the meditation literature, we hypothesized that there would be condition differences in EEG connectivity in the alpha and theta frequency bands. We further hypothesized that the subjective trance depth would positively relate to whole-brain connectivity, as is commonly observed in altered states of consciousness. Any finding from these analyses will inform the underlying mechanisms between the two externally silent but different internal psychological states.

2 Methods

2.1 Study overview

The data analyzed here were collected in a prior study ([Wahbeh et al., 2019](#)) and are briefly described below. We conducted a prospective within-subject design study with 13 adult trance channelers at the Institute of Noetic Sciences (IONS) between January 2018 and July 2018. EEG data were collected during two separate sessions of alternating trance channeling and mind-wandering states. The IONS Institutional Review Board (IRB) (IORG#0003743) approved all study activities, and the original study was pre-registered as a grant protocol (BIAL Foundation No 72/16).

2.2 Participants

The 13 participants were 57 ± 13 years old on average, mostly Caucasian (85%) and female (85%), educated (18 ± 4 years), and without any psychopathological diagnoses ([Wahbeh et al., 2019](#)). They met the inclusion and exclusion criteria, including being adults, self-defined full trance channelers, and being able to initiate connection at will and remain still while channeling. *Exclusions*: Significant acute

medical illness, psychotic symptoms, or dissociative symptoms. The validity of dis-carnate beings or whether the trance channeler is actually incorporating them is currently not definitively verifiable and, thus, was not focused on in the original study, nor will it be focused on in this study. The essential characteristic of the orig-inal study evaluated here is that the participants were proficient at entering trance states at will.

2.3 Study procedures

Participants were oriented, consented, and connected to electrophysiological monitoring devices while seated. Data collection occurred over two days using a counter-balanced design. Relevant to this study, each day’s data collection con-sisted of alternating 5-min trance and mind-wandering sessions with two 5-min breaks (Fig. 2).

2.4 Conditions

The research assistant alerted the participant to switch conditions and inserted an EEG signal marker simultaneously. The participant then inserted another marker with a keyboard button to indicate their transitions. Participants had as much time as necessary to transition between states.

Trance: The instructions for the trance state were for participants to enter their trance channeling state, as usual, staying as still and quiet as possible.

Mind-wandering condition: The instructions for the mind-wandering state were to actively think about one of three following things: (1) All the places you have traveled to or want to travel to in your life; (2) Walking through the grocery store for one of your regular shopping visits (i.e., imagine walking through aisles, picking out items, placing them in your cart, checking out); (3) Think about all the steps required to plant a garden and all the things you would plant. It should be noted that the mind-wandering con-dition used here resembles a mental imagery manipulation. True mind-wandering is challenging to induce in the laboratory by giving instructions because the act of the participant thinking in a particular way is not strictly mind-wandering.



FIG. 2 Laboratory session protocol. Conditions alternated between trance and mind-wandering with a counter-balanced design path A and path B. Wand = Mind wandering task. Trance = trance state (Wahbeh et al., 2019).

2.5 Subjective trance ratings

During the breaks, the research assistant would enter the electromagnetically shielded box and ask the participant, “How would you rate the level of consciousness during the last session?” anchored by 0 for “Fully conscious” to 11 for “Fully Unconscious.” The participant answered by marking on an 11 cm line. The research assistant measured the mark’s value and recorded it in the study database.

2.6 EEG preprocessing

64-channel EEG was recorded at 1024Hz with an ActiveTwo Biosemi system (Biosemi, Amsterdam, Netherlands). Please see [Wahbeh et al. \(2019\)](#), for detailed data collection methods. The data were reformatted to the brain imaging data structure (BIDS; [Cannard et al., 2022](#); [Pernet et al., 2019](#)) and imported into the EEGLAB software v2021b ([Delorme and Makeig, 2004](#)) in MATLAB (The Mathworks, Inc) using the *bids-matlab-tools* plugin v6.1. All further processing was done in EEGLAB. Note that older signal processing techniques were used in the original study (e.g., manual removal of abnormal channels and artifacts, manual tagging of bad independent components, etc.) that can lead to experimenter errors or simply missed artifacts (e.g., more subtle high-frequency artifacts). To address this issue and to facilitate replication, we only used automated open-source methods in the current study to process the data.

Data were averaged to reference, downsampled to 100Hz to increase computing speed, lowpass filtered at 45Hz with an elliptic infinite IIR filter (*iirfilt* v1.03 plugin, order=12, transition bandwidth 1 Hz), and highpass filtered with a non-causal FIR bandpass filter (*clean_rawdata* plugin v2.7; 0.25–1.25 Hz, order=252). Abnormal channels were automatically rejected by the *clean_rawdata* plugin (correlation threshold=0.75; other parameters set to default). For both conditions, data were segmented from 10s after the condition’s start to 1s before the condition’s end. Large abnormal EEG artifacts were then automatically rejected using the *clean_rawdata* plugin (BurstCriterion=20, other parameters set to default). Source separation was done using the preconditioned independent component analysis for real data (PICARD) plugin v1.0. (maximum iterations=500, data rank decreased by 1 to account for average reference), an algorithm for maximum likelihood ICA presenting state-of-the-art convergence speed ([Ablin et al., 2018](#)). Then, eye and muscle artifacts were automatically classified using the *ICLabel* plugin v1.3 (default parameters, i.e., 90% confidence of classification). Finally, bad channels were interpolated using spherical interpolation (*eeg_interp* function), and 2s adjacent non-overlapping segments were epoched (*eeg_regepoch* function).

2.7 Power spectra and connectivity

Source connectivity methods described below were performed using the *dipfit* plugin v4.3 and the *ROIconnect* plugin v1.0 ([Delorme, 2022](#)). Electrode locations were imported using the boundary element model (BEM) spherical coordinates.

The *leadfiled* matrix was computed using the Colin 27 MNI model (default parameters). Region of interest (ROI) activity was computed using the Desikan-Killiany surface atlas containing 68 regions. The signal source was localized and reconstructed using the linearly-constrained minimum variance (LCVM) beamforming default model with three principal components per region.

To compare results with the literature and source-reconstructed signal at the area level (better accounting for volume conduction effects), we estimated the power spectra at both the electrode and area levels using the EEGLAB *pop_precomp* function. The area-level spectral analysis offers a more refined estimation of the brain processes, which allows the determination of brain regions that might be involved in generating electrode power changes. Finally, connectivity examines how these areas interact and was estimated between pairs of regions using the cross-spectrum method, the fastest and simplest method providing the advantage of being less sensitive to volume conduction modeling errors. For all three measures, frequency bands were delineated for all analyses as follows: delta (1–3 Hz), theta (4–6 Hz), alpha (8–12 Hz), beta (18–22 Hz), and gamma (30–50 Hz).

2.8 Statistical analysis

2.8.1 Electrode- and area-level analyses

Parametric regression methods (e.g., linear regression) can lead to weak or biased model estimates when assumptions about linearity are not met, which is often the case in brain data. On the other hand, nonparametric regression methods account for nonlinearity and nonnormality and can allow the detection of patterns (e.g., nonlinear dynamics) that parametric techniques can miss. While most nonparametric techniques (e.g., kernel or smoothing splines) require large sample sizes, lack stability, and are very difficult to interpret because of the excessive transformations of the data, generalized additive models (GAMs) present solutions to these limitations while being able to model complex nonlinear patterns, many predictor variables, and to maintain interpretability.

Data manipulation and analysis were performed in the R programming language (R Development Core Team, 2021) to evaluate spectral differences between channeling and mind-wandering conditions at electrode and area levels. Spectral data were imported using the package *R.matlab* (Bengtsson et al., 2018) and transformed to long-form using the package *dplyr* (Wickham et al., 2022a) and *tidyr* (Wickham et al., 2022b). The following GAM was fit using the modified (i.e., *area_x*, *area_y*, *area_z*) function *bam()* in the package *mgcv* (Wood, 2011):

```
log(power) ~ condition + te(area_x, area_y, area_z, bs = "tp",
  by = condition, m = 2, k = c(15, 1515)) + s(subject, bs = "re")
+ s(segment, bs = "re") + s(session, bs = "re")
```

where areas were represented by their 3D spatial coordinates and smoothed via a thin plate regression spline for each frequency band. Note that smoothing accounted for spatial information across areas, which was not possible for the connectivity data as pairs have complex spatial information. Two model sets were fit: (1) with area corresponding to the 64 electrodes and (2) with areas corresponding to the 68 regions from the Desikan-Killiany surface atlas.

Basis dimensions and diagnostic plots were inspected for adherence to model assumptions. The difference between condition-specific smooths (specifically, the smooth for channeling and the smooth for mind-wandering) was computed using the approach described by [Rose and colleagues \(2012\)](#) with 2D smooths. To evaluate differences between conditions for each area, 95% confidence intervals and p -values were calculated, and P -values were adjusted using the False Discovery Rate (FDR) method ([Benjamini and Hochberg, 1995](#)). The GAM results were evaluated graphically using the package *ggplot2* ([Wickham, 2016](#)) and *ggsci* ([Xiao and Li, 2018](#)). The R package *gratia* was used to generate effects plots for the smooth terms in the model ([Simpson and Singmann, 2022](#)). Scalp topographies of the results were plotted with the *topoplot* function in EEGLAB ([Delorme and Makeig, 2004](#)), with the significant electrodes after correction displayed in red.

2.8.2 Connectivity-level analysis

The median connectivity (median across the six runs) was obtained for each subject and condition to conduct pairwise comparisons between the two conditions because the smoothing of the GAM was not feasible due to the complex spatial organization of connectivity between pairs of areas. Only the lower triangular was retained for analysis to avoid duplicate comparisons and unnecessary tests since connectivity data are stored as a symmetrical matrix.

The difference in connectivity between the two conditions was assessed using Yuen t -tests (paired t -tests on 20% trimmed means to deal with remaining outliers). Following the mass-univariate approach, the same pairwise comparisons were made under H_0 (the null hypothesis) using 1000-iterations bootstraps. The resulting t - and P -values were corrected for the type 1 error using the max and the spatiotemporal clustering methods ([Maris et al., 2007](#); [Pernet et al., 2011, 2015](#)). Since these advanced correction methods have not been adapted to connectivity data (two dimensions of areas as opposed to one dimension of electrodes), we adapted existing functions from the LIMO-EEG toolbox ([Pernet et al., 2015](#)) to apply spatiotemporal clustering with pairs of areas. The modified code can be found on Figshare with DOI 10.6084/m9.figshare.21084811.

2.8.3 Relationship between whole-brain connectivity and subjective trance ratings

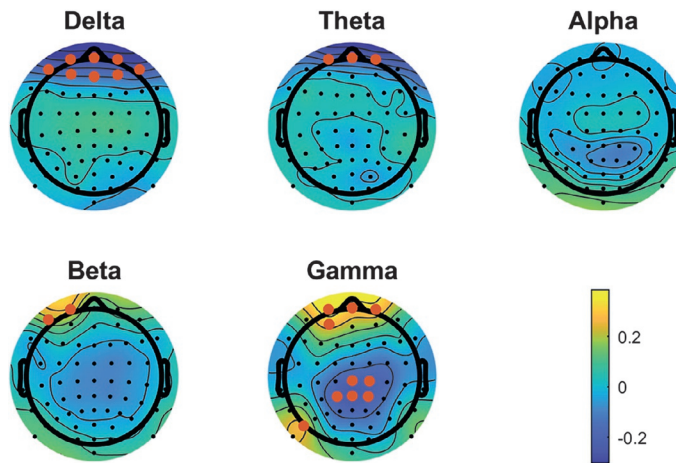
Connectivity values were averaged using 10% trimmed means to deal with outliers (very large and very small values) so that there was one value for each participant for each trance session (1–6; 3 on each day), resulting in 78 values (6 sessions $\times$ 13 participants) for each frequency band. Variables were assessed for normality

using the Shapiro-Wilk test for normality. The data were not normally distributed and resisted transformation to a normally distributed form. Differences between whole brain connectivity were evaluated with the nonparametric repeated measures ANOVA (Extended Mantel-Haenszel Stratified Test of Association). Nonparametric correlation analyses were conducted using Spearman's rank correlation tests to assess the relationship between subjective trance ratings (0=fully conscious to 11=fully unconscious) and whole-brain connectivity values. This correlation analysis was repeated with randomized mind-wandering trials to evaluate if correlation values were just due to chance. Please note that no subjective ratings were collected during the mind-wandering condition, and this step was used to assess if the correlations between subjective trance depth ratings and the trance whole-brain connectivity averages could be due to chance. The magnitude of the correlations was categorized based on recommendations by [Mukaka \(2012\)](#): Very high: 0.90–1.00; High: 0.70–0.89; Moderate: 0.50–0.69; Low: 0.30–0.49; Negligible: 0–0.30 ([Mukaka, 2012](#)). The resulting *P*-values were corrected for type 1 error using the FDR method.

3 Results

3.1 Spectral analysis

Differences between trance and mind-wandering survived multiple comparison correction ([Fig. 3](#)) for multiple electrodes. Eight frontal channels showed significantly greater delta power (1–3 Hz) during mind-wandering relative to trance: Fp1 (diff = -0.29 , $P = 0.00002$), AF7 (diff = -0.19 , $P = 0.03$), AF3 (diff = -0.14 , $P = 0.009$), Fpz (diff = -0.3 , $P = 0.00002$), Fp2 (diff = -0.31 , $P = 0.00002$), AF8 (diff = -0.23 , $P = 0.005$), AF4 (diff = -0.15 , $P = 0.008$), and AFz (diff = -0.13 , $P = 0.042$; all listed *P*'s are FDR-corrected; see FigShare 10.6084/m9.figshare.21084811 for statistical output). Three of these frontal channels also showed significantly greater theta power (4–6 Hz) during mind-wandering relative to trance: Fp1 (diff = -0.18 , $P = 0.021$), Fpz (diff = -0.19 , $P = 0.021$), and Fp2 (diff = -0.22 , $P = 0.005$). No difference was found between trance and mind-wandering at the electrode level for the alpha band (8–12 Hz; all *P*'s > 0.05). Two left frontal electrodes showed greater beta power during trance relative to mind-wandering: Fp1 (diff = 0.26 , $P = 0.000125$) and AF7 (diff = 0.19 , $P = 0.000125$). Four frontal and one left-temporal electrodes showed greater gamma power (30–50 Hz) during trance relative to mind-wandering: Fp1 (diff = 0.39 , $P = 0.0009$), Fpz (diff = 0.29 , $P = 0.015$), Fp2 (diff = 0.30 , $P = 0.015$), AF3 (diff = 0.28 , $P = 0.015$), and P7 (diff = 0.25 , $P = 0.017$). Finally, five centroparietal electrodes showed less gamma power during trance relative to mind-wandering: Cz (diff = -0.21 , $P = 0.042$), C2 (diff = -0.20 , $P = 0.042$), CP1 (diff = -0.21 , $P = 0.042$), CPz (diff = -0.24 , $P = 0.025$), and CP2 (diff = -0.21 , $P = 0.042$).

**FIG. 3**

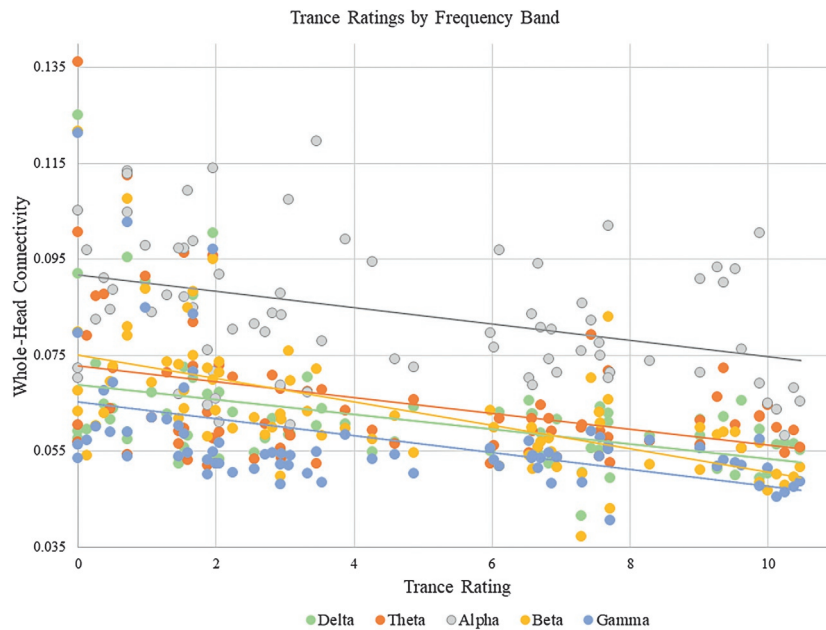
Spectral power differences between conditions at the electrode level by frequency band. Each scalp map represents (log-transformed) power spectral differences between the trance and mind-wandering conditions. Red dots denote electrodes with significant differences. No differences were observed in the alpha band. During mind-wandering relative to trance, greater delta and theta power was observed at the frontal electrodes, and greater gamma power at centro-parietal electrodes. During trance relative to mind-wandering, greater beta power was observed in the left frontal region and greater gamma power in frontal and temporal left regions.

No differences were found between the two conditions at the area-level (spectral and connectivity analyses; all P 's > 0.05 after multiple comparison corrections for both the max and the spatiotemporal clustering methods; see FigShare 10.6084/m9.figshare.21084811 for statistical output).

3.2 Subjective trance ratings and connectivity

As reported by [Wahbeh et al. \(2019\)](#), participants rated being in a deeper trance for subsequent sessions on each day (Day 1: Session 1 = 4.5 ± 3.0 , Session 2 = 5.0 ± 3.2 , Session 3 = 6.1 ± 3.4 ; Day 2: Session 1 = 5.2 ± 3.3 , Session 2 = 5.7 ± 3.6 ; Session 3 = 6.7 ± 3.6 ; $F(2,48) = 4.37$; $P = 0.018$). The depth of trance was not significantly different between the two days but marginally so ($F(1,24) = 4.04$; $P = 0.056$). There was no interaction between the day and the session.

Averaged whole brain connectivity was not different between the trance and mind-wandering states (all P 's > 0.05 ; see FigShare 10.6084/m9.figshare.21084811 for statistical output). Subjective trance ratings were negatively correlated with connectivity values such that reports of increased depth of trance (i.e., unconsciousness)

**FIG. 4**

Scatter plot showing the inverse correlation between subjective trance depth ratings and whole-brain (averaged) connectivity by frequency band (delta, theta, alpha, beta, and gamma). Each dot represents one participant's reported trance depth in relation to EEG connectivity during a trance session. Spearman correlations showed that subjective trance ratings were negatively correlated with whole-brain connectivity in all frequency bands, most strongly in beta (18–22 Hz) and gamma (30–50 Hz), followed by delta (1–3 Hz) and alpha (8–12 Hz), with theta (4–6 Hz) having the weakest relationship. Linear slopes are displayed for each frequency band.

corresponded with decreased whole-brain connectivity values in all frequency bands with multiple comparison correction using FDR (Fig. 4). Beta and gamma frequencies during channeling were moderately negatively correlated with subjective trance ratings (Beta $\rho^{\wedge} = -0.67$, $P = 0.000013$; Gamma $\rho^{\wedge} = -0.56$, $P < 0.000013$; corrected P -values are reported), and delta and alpha had low correlations (Delta $\rho^{\wedge} = -0.40$, $P = 0.0003$; Alpha $\rho^{\wedge} = -0.37$, $P = 0.001$). Theta was significant, with the weakest correlation strength (Theta $\rho^{\wedge} = -0.28$, $P = 0.013$). There were no significant correlations between trance ratings and the randomized mind-wandering power average data (ρ range from -0.09 to 0.26 , all P 's > 0.05 with FDR correction).

4 Discussion

In summary, these secondary analyses evaluated EEG spectral and connectivity differences between trance and mind-wandering states in thirteen participants who could enter trance states at will.

4.1 Spectral analyses at electrode and area-level

The electrode-level analysis showed significant differences between the two conditions.

There was less frontal delta power during trance compared to mind-wandering. The delta rhythm is dominant in young children and during motivated states (e.g., hunger, sexual arousal, and addiction; [Knyazev, 2012](#)). Some evidence suggests that delta oscillations reflect basic human functions such as cerebral spinal fluid flow (glymphatic system; [Benveniste et al., 2019](#)) or as a carrier of higher frequencies through long distances ([Knyazev, 2012](#)). This finding aligns with previous literature indicating increased broadband delta during mind-wandering compared to a breath-focus task ([Braboszcz and Delorme, 2011](#)). Generalized Delta may modulate sensory networks that are usually inactive to accomplish a specific mental task ([Harmony, 2013](#)). Hence, according to this interpretation, there could be a release of frontal inhibition during trance, which may potentially be required to go in and maintain a state of internal silence.

Similarly, we observed less frontal theta power (4–8 Hz) during trance relative to mind-wandering. Midline frontal theta oscillations are often linked to allocating and sustaining attention toward an object or task, as illustrated during mediation ([Brandmeyer and Delorme, 2020](#)) or during the detection of objects in space and spatial navigation ([Knyazev, 2007](#)). Hence, frontal theta activity is generally interpreted as an index of cognitive and attentional control ([Brandmeyer and Delorme, 2020](#); [Cavanagh and Frank, 2014](#); [Knyazev, 2007](#)). Thus, the current finding aligns with the literature showing decreased frontal theta power associated with increased activity of the default mode network (DMN; i.e., mind-wandering; [Brandmeyer and Delorme, 2020](#); [Scheeringa et al., 2008](#)). Thus, our finding further supports the idea that trance (i.e., internal silence) involves more internally-oriented attentional focus and cognitive control relative to mind-wandering.

No differences were observed in the alpha band (8–12 Hz). In general, alpha activity is the strongest EEG signal and is generally accepted to reflect the timing and gating of information across areas through top-down pulsed inhibition, presenting a potential mechanism for temporal attention (alpha synchronization and desynchronization; [Jensen and Mazaheri, 2010](#); [Klimesch et al., 2007](#); [Mathewson et al., 2011](#)). However, this is sometimes debated since contradictory results are found depending on which alpha sub-frequency, region, and participants are evaluated ([Nunez et al., 2001](#)). Others have suggested that solely evaluating alpha oscillations within a specific band is an incomplete assessment of alpha activity and that peak alpha frequency, phase, and variability of topographical and physiological factors should be incorporated into studies evaluating alpha EEG ([Bazanov and Vernon, 2014](#)). These factors may explain the mixed results evaluating alpha activity during trance compared to control conditions, with most studies finding increased alpha ([Bastos et al., 2015, 2016](#); [Hughes, 1991](#); [Kawai et al., 2017](#); [Oohashi et al., 2002](#)) and others finding no difference ([Sabourin et al., 1990](#)). Future studies could incorporate these additional components to evaluate alpha's role in trance and mental silence further.

Beta (13–32 Hz) oscillations were increased during trance in two left frontal electrodes. This finding aligns with other studies demonstrating that decreases in beta

power (especially in concert with increases in theta) can indicate mind-wandering (van Son et al., 2019). Frontal beta oscillations have also been associated with top-down control and regulation (Stoll et al., 2016). Interestingly, increased right frontal beta power has been linked to preventing a thought from coming to awareness, a process that can be involved in maintaining mental silence (Castiglione et al., 2019; Wagner et al., 2018).

Finally, four frontal electrodes and one parietal electrode had greater gamma oscillations during trance than mind-wandering. Other studies have also observed increased gamma during trance states (symmetrical parietal region in Flor-Henry et al., 2017; right frontotemporal region in Lehmann et al., 2001). Furthermore, meditation induces synchronization and increased amplitude in gamma band activity in frontoparietal areas (Lutz et al., 2004) and parietal-occipital regions (Braboszcz et al., 2017). While findings in the gamma frequencies are often debated because they are prone to contain high-frequency artifacts that are hard to detect and remove, gamma oscillations are generally associated with object representation and entail the binding of separate parts of the same object (Tallon-Baudry and Bertrand, 1999). This is done through bottom-up processes and the activation, retrieval, or rehearsal of an internal representation through a top-down process, integration of sensory and motor processes during movement (Tallon-Baudry and Bertrand, 1999). Gamma activity is associated with local functional connectivity, or small brain networks, which then connect to global networks by slow frequencies (Başar-Eroglu et al., 1996). Because gamma band power increases during complex and attention-demanding tasks, induced gamma activity is often interpreted as the neural substrate of cognitive processes (Başar-Eroglu et al., 1996). More specifically, increased frontal gamma is related to the destabilization of the perceptual system when viewing ambiguous patterns that induce multistable perception (Başar-Eroglu et al., 1996). Interestingly, we see lower gamma during trance in central-parietal electrodes. Increases in gamma oscillations in central electrodes are associated with self-referential tasks compared to other-judging tasks (Mu and Han, 2010). These results align with the subjective reports of loss of sense of self during trance states (Cardena et al., 2009).

These significant results differ from the non-significant electrode level analyses conducted in the original study (Wahbeh et al., 2019). The dissimilar results are likely a result of different signal processing methods employed (advanced, validated, automated algorithms to extract EEG artifacts from the signal in this study as opposed to manual cleaning and older ICA methods) and statistical testing employed (GAM here to better capture nonlinear complexity in the signal and account for covariates).

Since scalp-level analyses can be biased by volume conduction effects (see Section 1), we conducted a novel inverse source method to evaluate the power spectra at the area level. When the source localization and reconstruction were applied, no spectral differences were observed between the two conditions in the 68 brain areas. We can only surmise why this is the case. This might be explained by the fact that this method is ultimately an approximation, and some researchers

argue that electrode-level data might still be more reliable because it is not as transformed. Electrode data was used to reconstruct activity into dipoles, grouped by brain area, and subjected to principal component analysis (PCA) to generate the area-level data. Each brain area corresponds to a time series of multiple voxels, and the PCA reduces each area to three components. In selecting three, others are discarded, usually around 10% of the information. If the significant difference is contained within this 10% or if the differences in the electrode level data are segmented into separate brain areas, then any difference between conditions would no longer remain. Hence, it is possible that the transformations involved excluded data that drove the difference between the two conditions. Furthermore, generic source localization approximations using brain atlas templates may not isolate the brain area well for specific individuals due to differences in brain anatomy across individuals.

Additionally, EEG data were averaged to reference during processing, highlighting global, widespread scalp dynamics and greatly reducing local dynamics (Nunez et al., 2016; Nunez and Srinivasan, 2006). This can be solved using reference-free Surface Laplacian transformation (also called current-source-density transformation; Nunez et al., 2016; Nunez and Srinivasan, 2006). Thus, future studies might better capture local/global dynamics using this transformation type by providing more accurate source separation and spatial accuracy. These limitations also apply to the absence of findings in the connectivity measure.

Overall, these findings highlight the importance of EEG processing methods and their influence on results and should be considered when interpreting the results of EEG studies. Further validation of our findings should be assessed in future studies. In addition, fully transparent data and processing pipelines allow for the verification of analyses and replication. All data and code are available at Figshare with DOI 10.6084/m9.figshare.21084811.

4.2 Connectivity

Functional connectivity evaluating how brain areas interact was estimated, and contrary to our expectations, we found no differences in connectivity between the mind-wandering and trance states. It is conceivable that these source-reconstructed measures might simply not capture these large differences in subjective experience. Other novel nonlinear measures (e.g., entropy, chaos, fractal) have shown promising results in discriminating between consciousness states (e.g., coma; Eagleman et al., 2019) and may be better suited to study trance states.

However, subjective trance depth ratings were significantly correlated to whole-brain connectivity (i.e., connectivity values averaged across all area pairs) in all frequency bands. The deeper the trance, the less overall functional connectivity was observed in the brain overall. Since connectivity was estimated using the cross-spectrum method from source-reconstructed signals, a simplistic interpretation of reduced overall connectivity is that brain areas were doing different and more localized “things” to a greater extent during trance than during mind-wandering. In other words,

there was less widespread synchrony and coherence between close and distant brain areas during trance states. This relationship was most robust for beta, followed by gamma, delta, alpha, and finally, theta, with the weakest relationship, confirming the tendency for high-frequency activity to be local and slow frequencies to be more present in long-distance connections.

These relationships between phenomenological states and whole-brain EEG connectivity are exploratory, although other neuroimaging studies have observed reduced connectivity in trance or trance-related states. For example, one group found underactivation in the left culmen, left hippocampus, left inferior occipital gyrus, left anterior cingulate, right superior temporal gyrus, and right precentral gyrus during trance compared to non-trance state using single-photon emission computed tomography (Peres et al., 2012). Another group using fMRI found decreased connectivity between the executive control network and default mode network (DMN) and increased connectivity between the executive control and the salience network during hypnotic trance compared to a rest condition (Jiang et al., 2017). The salience network is thought to modulate resources toward either the DMN (during rest, mind-wandering) or the executive control network (task, focusing attention; (Menon and Uddin, 2010). However, it is important to note that while the DMN is generally considered to be anticorrelated with task-related networks, such as attentional and executive (Mittner et al., 2016), mind-wandering has now been found to display increased connectivity within the DMN and between the DMN and other networks, such as large-scale networks, including attentional and executive control ones (Godwin et al., 2017; Kucyi, 2018; Mittner et al., 2016; Mooneyham et al., 2017; Zhou and Lei, 2018). Thus, our findings at the electrode and the whole-brain connectivity levels align with other studies demonstrating generally increased functional connectivity between broad networks in lower frequencies (delta, theta) during mind-wandering, with more localized, asynchronous activity during trance in higher frequencies (beta, gamma).

Other studies have evaluated the relationship between alpha connectivity and unconsciousness. For example, in studies with ketamine, a commonly used anesthetic, researchers found decreased connectivity in alpha and the frontoparietal network (Blain-Moraes et al., 2014). These results mirror what is observed with propofol, another commonly used anesthetic (Lee et al., 2013). The authors remarked that their results “suggest that directional connectivity changes in the alpha bandwidth may be state-related markers of unconsciousness” (Blain-Moraes et al., 2014).

As far as we know, this is the first study to evaluate participants’ self-report experience of trance depth and whole-brain EEG connectivity. While it is a small first step in evaluating these relationships, the greater ease with which EEG can be collected compared to other neuroimaging methods that are more costly and invasive opens the door to expanded research into this area. The methods used to calculate connectivity with EEG data are novel and made available to the neuroscience community, accelerating research in this direction. Using open-source code with EEGLAB makes replicating this study and using this connectivity measure in other studies possible.

4.3 Limitations

Several limitations should be considered when reviewing this study's results. Primarily, these analyses are secondary to a pre-registered formal evaluation. The results are exploratory and should be viewed as a starting point for future confirmatory analyses. Implicit in the study's exploratory nature is that the data collection did not consider these future analyses. For example, we did not collect subjective ratings during the mind-wandering condition, only the trance condition. We assumed that the participants were mind-wandering (i.e., following task instructions) during that condition and not in a trance state but did not follow up to confirm that this was so. In addition, we assumed that the trance state supports mental silence. This assumption is based on the literature and anecdotal reports from channelers who report mental silence while in a trance. However, we did not specifically ask the participants if they were experiencing mental silence during the trance condition. Future research can address this by implementing methods to ascertain if mental silence occurs during trance states. We can learn from meditation studies evaluating transcendence in planning future studies, incorporating methods such as retrospective recall, eye blinks, bell tones at specific intervals, button presses, or another marker/note when entering or leaving the mentally silent state ([Wahbeh et al., 2018b](#)).

Another limitation is the inherent ambiguity of source localization methods without a definitive solution to the inverse problem. While these methods are now more robustly validated, we can not be certain that the brain areas arising from the source localization analysis definitively correspond to those areas in the participants.

Another possible improvement in future studies could be the pre-selection of clusters of areas known to be connected anatomically or functionally to reduce the high number of statistical tests (i.e., 4624 pairwise comparisons). Differences observed at the whole-brain level may be visible by grouping some area pairs into clusters without losing information. This approach is commonly used in other fields of research ([Deshpande and Jia, 2020](#); [Shakil et al., 2014](#)).

In addition, the participant number was small, demographically homogenous, and limited to one trance induction type. Thus, generalizability to the general population and other trance induction techniques is precluded. Future studies could incorporate more diverse participants and use various trance induction techniques to explore any differences between these two potentially influencing factors.

5 Conclusion

We evaluated EEG spectral and connectivity differences between trance and mind-wandering states. We observed significantly increased power during mind-wandering in delta and theta frequency bands and during trance in beta and gamma frequency bands in frontal electrodes. In addition, gamma power was lower during trance for central and central-parietal electrodes. Connectivity was not different between the trance and mind-wandering states. However, there was a relationship

between subjective ratings of trance depth and connectivity, with increased trance depth reflecting decreased whole-brain connectivity in all frequency bands. Trance offers an opportunity to enter mentally silent states and explore neurophenomenological relationships further. Future studies should include self-report assessment of trance depth and level of mental silence.

Acknowledgments

The authors would like to thank the IONS Science team, IONS staff and members, the study participants, the Ray Benton Foundation, the Federico and Elvia Faggin Foundation, and FUNDAÇÃO BIAL (Grant 72/16) for their support of this and other projects in this research program. The authors would also like to thank [Maria Wojakowski](#) for her statistical support of this project.

Conflict of interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Author contributions

Helané Wahbeh – Conceptualization, funding acquisition, project administration, writing – original draft preparation, review & editing.

Cedric Cannard – Conceptualization, writing review & editing.

Arnaud Delorme – Conceptualization, funding acquisition, writing – original draft preparation, review & editing.

Funding

This work was supported by the Ray Benton Foundation, the Federico and Elvia Faggin Foundation and FUNDAÇÃO BIAL (Grant 72/16), the Institute of Noetic Sciences, and its members. The funders did not play a role in the design or implementation of the present study.

References

- Ablin, P., Cardoso, J.-F., Gramfort, A., 2018. Faster independent component analysis by pre-conditioning with Hessian approximations. *IEEE Trans. Signal Process.* 66, 4040–4049. <https://doi.org/10.1109/TSP.2018.2844203>.
- Alderson-Day, B., Fernyhough, C., 2015. Inner speech: development, cognitive functions, phenomenology, and neurobiology. *Psychol. Bull.* 141, 931–965. <https://doi.org/10.1037/bul0000021>.

- American Psychological Association, 2022. Altered States of Consciousness (ASC) [WWW Document]. APA Dictionary of Psychology. <https://dictionary.apa.org/altered-state-of-consciousness>. (accessed 11.11.22).
- Baillet, S., Riera, J.J., Marin, G., Mangin, J.F., Aubert, J., Garnero, L., 2001. Evaluation of inverse methods and head models for EEG source localization using a human skull phantom. *Phys. Med. Biol.* 46, 77–96. <https://doi.org/10.1088/0031-9155/46/1/306>.
- Baird, B., Smallwood, J., Mrazek, M.D., Kam, J.W.Y., Franklin, M.S., Schooler, J.W., 2012. Inspired by distraction: mind wandering facilitates creative incubation. *Psychol. Sci.* 23, 1117–1122. <https://doi.org/10.1177/0956797612446024>.
- Başar-Eroglu, C., Strüber, D., Kruse, P., Başar, E., Stadler, M., 1996. Frontal gamma-band enhancement during multistable visual perception. *Int. J. Psychophysiol. New Adv. EEG Cogn.* 24, 113–125. [https://doi.org/10.1016/S0167-8760\(96\)00055-4](https://doi.org/10.1016/S0167-8760(96)00055-4).
- Bastos, M.A.V., Bastos, P.R.H., Goncalves, L.M., Osorio, I.H.S., Lucchetti, G., 2015. Mediumship: review of quantitative studies published in the 21st century. *Arch. Clin. Psychiatry* 42, 129–138.
- Bastos, M.A.V., Bastos, P.R.H., Osório, I.H.S., Muass, K.A.R.C., Iandoli Jr., D., Lucchetti, G., 2016. Frontal electroencephalographic (EEG) activity and mediumship: a comparative study between spiritist mediums and controls. *Arch. Clin. Psychiatr. (São Paulo)* 43, 20–26. <https://doi.org/10.1590/0101-60830000000076>.
- Bazanov, O.M., Vernon, D., 2014. Interpreting EEG alpha activity. *Neurosci. Biobehav. Rev.* 44, 94–110. <https://doi.org/10.1016/j.neubiorev.2013.05.007>. Applied Neuroscience: Models, methods, theories, reviews. A Society of Applied Neuroscience (SAN) special issue.
- Bengtsson, H., Jacobson, A., Reidy, J., 2018. R.matlab: Read and Write MAT Files and Call MATLAB from Within R.
- Benjamini, Y., Hochberg, Y., 1995. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *J. R. Stat. Soc. Ser. B* 57, 289–300. <https://doi.org/10.1111/j.2517-6161.1995.tb02031.x>.
- Ben-Soussan, T.D., Mauro, F., Lasaponara, S., Glicksohn, J., Marson, F., Berkovich-Ohana, A., 2019. Chapter 7—Fully immersed: state absorption and electrophysiological effects of the OVO whole-body perceptual deprivation chamber. In: Srinivasan, N. (Ed.), *Progress in Brain Research. Elsevier, Meditation*, pp. 165–184. <https://doi.org/10.1016/bs.pbr.2018.10.023>.
- Benveniste, H., Liu, X., Koundal, S., Sanggaard, S., Lee, H., Wardlaw, J., 2019. The glymphatic system and waste clearance with brain aging: a review. *GER* 65, 106–119. <https://doi.org/10.1159/000490349>.
- Blain-Moraes, S., Lee, U., Ku, S., Noh, G., Mashour, G.A., 2014. Electroencephalographic effects of ketamine on power, cross-frequency coupling, and connectivity in the alpha bandwidth. *Front. Syst. Neurosci.* 8.
- Braboszcz, C., Delorme, A., 2011. Lost in thoughts: neural markers of low alertness during mind wandering. *Neuroimage* 54, 3040–3047.
- Braboszcz, C., Cahn, B.R., Levy, J., Fernandez, M., Delorme, A., 2017. Increased gamma brainwave amplitude compared to control in three different meditation traditions. *PLoS One* 12, e0170647. <https://doi.org/10.1371/journal.pone.0170647>.
- Brandmeyer, T., Delorme, A., 2020. Closed-loop frontal midline neurofeedback: a novel approach for training focused-attention meditation. *Front. Hum. Neurosci.* 14. <https://doi.org/10.3389/fnhum.2020.00246>.

- Brandmeyer, T., Delorme, A., 2021. Meditation and the wandering mind: a theoretical framework of underlying neurocognitive mechanisms. *Perspect. Psychol. Sci.* 16, 39–66. <https://doi.org/10.1177/1745691620917340>.
- Brookes, M.J., Hale, J.R., Zumer, J.M., Stevenson, C.M., Francis, S.T., Barnes, G.R., Owen, J.P., Morris, P.G., Nagarajan, S.S., 2011. Measuring functional connectivity using MEG: methodology and comparison with fMRI. *Neuroimage* 56, 1082–1104. <https://doi.org/10.1016/j.neuroimage.2011.02.054>.
- Buhrman, S., 1997. Trance types and amnesia revisited: using detailed interviews to fill in the gaps. *Anthropol. Conscious.* 8, 10–21. <https://doi.org/10.1525/ac.1997.8.1.10>.
- Cahn, R., Polich, J., 2006. Meditation states and traits: EEG, ERP, and neuroimaging studies. *Psychol. Bull.* 132, 180–211.
- Cannard, C., Wabbeh, H., Delorme, A., 2022. Trance channeling EEG study—OpenNeuro. Open Neuro. <https://doi.org/10.18112/openneuro.ds004040.v1.0.0>.
- Cardeña, E., Van Duijl, M., Weiner, L., Terhune, D., 2009. Possession/Trance Phenomena. *Dissociation and the Dissociative Disorders: DSM-V and Beyond*. pp. 171–181.
- Carmody, J., Baer, R.A., 2008. Relationships between mindfulness practice and levels of mindfulness, medical and psychological symptoms and well-being in a mindfulness-based stress reduction program. *J. Behav. Med.* 31, 23–33. <https://doi.org/10.1007/s10865-007-9130-7>.
- Castiglione, A., Wagner, J., Anderson, M., Aron, A.R., 2019. Preventing a thought from coming to mind elicits increased right frontal beta just as stopping action does. *Cereb. Cortex* 29, 2160–2172. <https://doi.org/10.1093/cercor/bhz017>.
- Castillo, R.J., 1995. Culture, trance, and the mind-brain. *Anthropol. Conscious.* 6, 17–34.
- Castillo, R.J., 2003. Trance, functional psychosis, and culture. *Psychiatr. Interpers. Biol. Process.* 66, 9–21.
- Cavanagh, J.F., Frank, M.J., 2014. Frontal theta as a mechanism for cognitive control. *Trends Cogn. Sci.* 18, 414–421. <https://doi.org/10.1016/j.tics.2014.04.012>.
- Chhaya, L., 2013. World's quietest room: anechoic chambers for EMC testing. *i-Manage. J. Electron. Eng.* 3, 1.
- Cox, T., 2014. *The Sound Book: The Science of the Sonic Wonders of the World*. W.W Norton & Company.
- Delorme, A., 2022. ROI Connectivity Toolbox for EEGLAB.
- Delorme, A., Brandmeyer, T., 2019. When the meditating mind wanders. *Curr. Opin. Psychol.* 28, 133–137. <https://doi.org/10.1016/j.copsyc.2018.12.006>.
- Delorme, A., Makeig, S., 2004. EEGLAB: an open source toolbox for analysis of single-trial EEG dynamics. *J. Neurosci. Methods* 134, 9–21.
- Deshpande, G., Jia, H., 2020. Multi-level clustering of dynamic directional brain network patterns and their behavioral relevance. *Front. Neurosci.* 13.
- Ding, L., He, B., 2008. Sparse source imaging in electroencephalography with accurate field modeling. *Hum. Brain Mapp.* 29 (9), 1053–1067.
- Dodich, A., Zollo, M., Crespi, C., Cappa, S.F., Laureiro Martinez, D., Falini, A., Canessa, N., 2019. Short-term Sahaja Yoga meditation training modulates brain structure and spontaneous activity in the executive control network. *Brain Behav.* 9, e01159. <https://doi.org/10.1002/brb3.1159>.
- Domingo-Pueyo, A., Sanz-Valero, J., Wanden-Berghe, C., 2016. Disorders induced by direct occupational exposure to noise: systematic review. *Noise Health* 18, 229–239. <https://doi.org/10.4103/1463-1741.192479>.

- Eagleman, S.L., Chander, D., Reynolds, C., Ouellette, N.T., MacIver, M.B., 2019. Nonlinear dynamics captures brain states at different levels of consciousness in patients anesthetized with propofol. *PLoS One* 14, e0223921. <https://doi.org/10.1371/journal.pone.0223921>.
- Fancourt, D., Ockelford, A., Belai, A., 2014. The psychoneuroimmunological effects of music: a systematic review and a new model. *Brain, Behavior, and Immunity* 36, 15–26. <https://doi.org/10.1016/j.bbi.2013.10.014>.
- Flor-Henry, P., Shapiro, Y., Sombrun, C., 2017. Brain changes during a shamanic trance: altered modes of consciousness, hemispheric laterality, and systemic psychobiology. *Cogent Psychol.* 4, 1313522. <https://doi.org/10.1080/23311908.2017.1313522>.
- Franklin, M., Mrazek, M., Anderson, C., Smallwood, J., Kingstone, A., Schooler, J., 2013. The silver lining of a mind in the clouds: interesting musings are associated with positive mood while mind-wandering. *Front. Psychol.* 4. <https://www.frontiersin.org/articles/10.3389/fpsyg.2013.00583>.
- Glicksohn, J., Berkovich-Ohana, A., Mauro, F., Ben-Soussan, T.D., 2017. Time perception and the experience of time when immersed in an altered sensory environment. *Front. Hum. Neurosci.* 11.
- Godwin, C.A., Hunter, M.A., Bezdek, M.A., Lieberman, G., Elkin-Frankston, S., Romero, V.L., Witkiewitz, K., Clark, V.P., Schumacher, E.H., 2017. Functional connectivity within and between intrinsic brain networks correlates with trait mind wandering. *Neuropsychologia* 103, 140–153. <https://doi.org/10.1016/j.neuropsychologia.2017.07.006>.
- Gross, J., Kujala, J., Hämäläinen, M., Timmermann, L., Schnitzler, A., Salmelin, R., 2001. Dynamic imaging of coherent sources: studying neural interactions in the human brain. *Proc. Natl. Acad. Sci. U.S.A.* 98 (2), 694–699.
- Grova, C., Daunizeau, J., Kobayashi, E., Bagshaw, A.P., Lina, J.-M., Dubeau, F., Gotman, J., 2008. Concordance between distributed EEG source localization and simultaneous EEG-fMRI studies of epileptic spikes. *Neuroimage* 39, 755–774. <https://doi.org/10.1016/j.neuroimage.2007.08.020>.
- Gwilliams, L., King, J.-R., Marantz, A., Poeppel, D., 2022. Neural dynamics of phoneme sequences reveal position-invariant code for content and order. *Nat. Commun.* 13, 6606. <https://doi.org/10.1038/s41467-022-34326-1>.
- Harmony, T., 2013. The functional significance of delta oscillations in cognitive processing. *Front. Integr. Neurosci.* 7.
- Haufe, S., 2015. An extendable simulation framework for benchmarking EEG-based brain connectivity estimation methodologies. In: 2015 37th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC). Presented at the 2015 37th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), pp. 7562–7565. <https://doi.org/10.1109/EMBC.2015.7320142>.
- Haufe, S., Ewald, A., 2019. A simulation framework for benchmarking EEG-based brain connectivity estimation methodologies. *Brain Topogr* 32, 625–642. <https://doi.org/10.1007/s10548-016-0498-y>.
- Haufe, S., Nikulin, V.V., Müller, K.-R., Nolte, G., 2013. A critical assessment of connectivity measures for EEG data: a simulation study. *Neuroimage* 64, 120–133. <https://doi.org/10.1016/j.neuroimage.2012.09.036>.
- Haufe, S., Nikulin, V.V., Ziehe, A., Müller, K.R., Nolte, G., 2008. Combining sparsity and rotational invariance in EEG/MEG source reconstruction. *NeuroImage* 42 (2), 726–738.
- Haufe, S., Treder, M.S., Gugler, M.F., Sagebaum, M., Curio, G., Blankertz, B., 2011. EEG potentials predict upcoming emergency brakings during simulated driving. *J. Neural Eng.* 8 (5), 056001.

- Hendriks, T., 2018. The effects of Sahaja Yoga meditation on mental health: a systematic review. *J. Complement. Integr. Med.* 15. <https://doi.org/10.1515/jcim-2016-0163>.
- Hernández, S.E., Suero, J., Rubia, K., González-Mora, J.L., 2015. Monitoring the neural activity of the state of mental silence while practicing Sahaja Yoga meditation. *J. Altern. Complement. Med.* 21, 175–179. <https://doi.org/10.1089/acm.2013.0450>.
- Hernández, S.E., Barros-Loscertales, A., Xiao, Y., González-Mora, J.L., Rubia, K., 2018. Gray matter and functional connectivity in anterior cingulate cortex are associated with the state of mental silence during Sahaja Yoga meditation. *Neuroscience* 371, 395–406. <https://doi.org/10.1016/j.neuroscience.2017.12.017>.
- Hettich, D.T., Bolinger, E., Matuz, T., Birbaumer, N., Rosenstiel, W., Spüler, M., 2016. EEG responses to auditory stimuli for automatic affect recognition. *Front. Neurosci.* 10.
- Hinterberger, T., Schmidt, S., Kamei, T., Walach, H., 2014. Decreased electrophysiological activity represents the conscious state of emptiness in meditation. *Front. Psychol.* 5, 99. <https://doi.org/10.3389/fpsyg.2014.00099>.
- Huang, Y., Parra, L.C., Haufe, S., 2016. The New York head—a precise standardized volume conductor model for EEG source localization and tES targeting. *NeuroImage* 140, 150–162. <https://doi.org/10.1016/j.neuroimage.2015.12.019>.
- Hughes, D.J., 1991. Blending with an other: an analysis of trance channeling in the United States. *Ethos* 19, 161–184.
- Jacobs, T.L., Epel, E.S., Lin, J., Blackburn, E.H., Wolkowitz, O.M., Bridwell, D.A., Zanesco, A.P., Aichele, S.R., Sahdra, B.K., MacLean, K.A., King, B.G., Shaver, P.R., Rosenberg, E.L., Ferrer, E., Wallace, B.A., Saron, C.D., 2011. Intensive meditation training, immune cell telomerase activity, and psychological mediators. *Psychoneuroendocrinology* 21035949 (36), 664–681. <https://doi.org/10.1016/j.psyneuen.2010.09.010>.
- Jensen, O., Mazaheri, A., 2010. Shaping functional architecture by oscillatory alpha activity: gating by inhibition. *Front. Hum. Neurosci.* 4, 186. <https://doi.org/10.3389/fnhum.2010.00186>.
- Jiang, H., White, M.P., Greicius, M.D., Waelde, L.C., Spiegel, D., 2017. Brain activity and functional connectivity associated with hypnosis. *Cereb. Cortex* 27, 4083–4093. <https://doi.org/10.1093/cercor/bhw220>.
- Jonmohamadi, Y., Poudel, G., Innes, C., Weiss, D., Krueger, R., Jones, R., 2014. Comparison of beamformers for EEG source signal reconstruction. *Biomed. Sig. Process. Control* 14, 175–188. <https://doi.org/10.1016/j.bspc.2014.07.014>.
- Kamiński, M.J., Blinowska, K.J., 1991. A new method of the description of the information flow in the brain structures. *Biol. Cybern.* 65, 203–210. <https://doi.org/10.1007/BF00198091>.
- Kawai, N., Honda, M., Nishina, E., Yagi, R., Oohashi, T., 2017. Electroencephalogram characteristics during possession trances in healthy individuals. *Neuroreport* 28, 949.
- Kiebel, S.J., Garrido, M.I., Moran, R.J., Friston, K.J., 2008. Dynamic causal modelling for EEG and MEG. *Cogn Neurodyn* 2, 121–136. <https://doi.org/10.1007/s11571-008-9038-0>.
- Killingsworth, M.A., Gilbert, D.T., 2010. A wandering mind is an unhappy mind. *Science* 330, 932. <https://doi.org/10.1126/science.1192439>.
- Klimesch, W., Sauseng, P., Hanslmayr, S., 2007. EEG alpha oscillations: the inhibition-timing hypothesis. *Brain Res. Rev.* 16887192 (53), 63–88.
- Knyazev, G.G., 2007. Motivation, emotion, and their inhibitory control mirrored in brain oscillations. *Neurosci. Biobehav. Rev.* 31, 377–395.
- Knyazev, G.G., 2012. EEG delta oscillations as a correlate of basic homeostatic and motivational processes. *Neurosci. Biobehav. Rev.* 36, 677–695. <https://doi.org/10.1016/j.neubiorev.2011.10.002>.

- Kucyi, A., 2018. Just a thought: how mind-wandering is represented in dynamic brain connectivity. *NeuroImage* 180, 505–514. <https://doi.org/10.1016/j.neuroimage.2017.07.001>.
- Lee, H., Mashour, G.A., Noh, G.-J., Kim, S., Lee, U., 2013. Reconfiguration of network hub structure after propofol-induced unconsciousness. *Anesthesiology* 119, 1347–1359. <https://doi.org/10.1097/ALN.0b013e3182a8ec8c>.
- Lehmann, D., Faber, P.L., Achermann, P., Jeanmonod, D., Gianotti, L.R., Pizzagalli, D., 2001. Brain sources of EEG gamma frequency during volitionally meditation-induced, altered states of consciousness, and experience of the self. *Psychiatry Res.* 11738545 (108), 111–121.
- Lilly, J.G., 2019. Designing the quietest room in the world. *J. Acoust. Soc. Am.* 146, 2767. <https://doi.org/10.1121/1.5136582>.
- Lutz, A., Greischar, L.L., Rawlings, N.B., Ricard, M., Davidson, R.J., 2004. Long-term meditators self-induce high-amplitude gamma synchrony during mental practice. *Proc. Natl. Acad. Sci. U.S.A.* 15534199 (101), 16369–16373.
- Mainieri, A.G., Peres, J.F.P., Moreira-Almeida, A., Mathiak, K., Habel, U., Kohn, N., 2017. Neural correlates of psychotic-like experiences during spiritual-trance state. *Psychiatr. Res. Neuroimaging* 266, 101–107. <https://doi.org/10.1016/j.psychresns.2017.06.006>.
- Manocha, R., Semmar, B., Black, D., 2007. A pilot study of a mental silence form of meditation for women in perimenopause. *J. Clin. Psychol. Med. Settings* 14, 266–273. <https://doi.org/10.1007/s10880-007-9076-5>.
- Manocha, R., Black, D., Sarris, J., Stough, C., 2011. A randomized, controlled trial of meditation for work stress, anxiety and depressed mood in full-time workers. *Evid. Based Complement. Alternat. Med.* 2011, e960583. <https://doi.org/10.1155/2011/960583>.
- Maris, E., Schoffelen, J.-M., Fries, P., 2007. Nonparametric statistical testing of coherence differences. *J. Neurosci. Methods* 163, 161–175.
- Mason, O.J., Brady, F., 2009. The psychotomimetic effects of short-term sensory deprivation. *J. Nerv. Mental Dis.* 197, 783–785. <https://doi.org/10.1097/NMD.0b013e3181b9760b>.
- Mathewson, K.E., Lleras, A., Beck, D.M., Fabiani, M., Ro, T., Gratton, G., 2011. Pulsed out of awareness: EEG alpha oscillations represent a pulsed-inhibition of ongoing cortical processing. *Front. Psychol.* 2, 99. <https://doi.org/10.3389/fpsyg.2011.00099>.
- Menon, V., Uddin, L.Q., 2010. Saliency, switching, attention and control: a network model of insula function. *Brain Struct. Funct.* 214, 655–667. <https://doi.org/10.1007/s00429-010-0262-0>.
- Mittner, M., Hawkins, G.E., Boekel, W., Forstmann, B.U., 2016. A neural model of mind wandering. *Trends Cogn. Sci.* 20, 570–578. <https://doi.org/10.1016/j.tics.2016.06.004>.
- Mooneyham, B.W., Schooler, J.W., 2013. The costs and benefits of mind-wandering: a review. *Can. J. Exp. Psychol.* 67, 11–18. <https://doi.org/10.1037/a0031569>.
- Mooneyham, B., Schooler, J., 2015. Mind wandering and meta-awareness in hypnosis and meditation relating executive function across states of consciousness. In: *Hypnosis and Meditation: Towards an Integrative Science of Conscious Planes*, pp. 221–240.
- Mooneyham, B., Abrams, M., Mrazek, A., Mrazek, K., Phillips, D., Schooler, J., 2017. States of mind: characterizing the neural bases of focus and mind-wandering through dynamic functional connectivity. *J. Cogn. Neurosci.* 29, 495–506. https://doi.org/10.1162/jocn_a_01066.
- Mrazek, M.D., Franklin, M.S., Phillips, D.T., Baird, B., Schooler, J.W., 2013. Mindfulness training improves working memory capacity and GRE performance while reducing mind wandering. *Psychol. Sci.* 24, 776–781. <https://doi.org/10.1177/0956797612459659>.
- Mu, Y., Han, S., 2010. Neural oscillations involved in self-referential processing. *NeuroImage* 53, 757–768. <https://doi.org/10.1016/j.neuroimage.2010.07.008>.

- Mukaka, M., 2012. A guide to appropriate use of correlation coefficient in medical research. *Malawi Med. J.* 24, 69–71.
- Nave, C.R., 2016. Threshold of Hearing [WWW Document]. HyperPhysics. Department of Physics and Astronomy, Georgia State University. URL <http://hyperphysics.phy-astr.gsu.edu/hbase/Sound/earcrrv.html> (accessed 7.21.22).
- Negro Jr., P.J., Palladino-Negro, P., Louzã, M.R., 2002. Do religious mediumship dissociative experiences conform to the sociocognitive theory of dissociation? *J. Trauma Dissociation* 3, 51–73.
- Nunez, M.D., Nunez, P.L., Srinivasan, R., Ombao, H., Linquist, M., Thompson, W., et al., 2016. Electroencephalography (EEG): neurophysics, experimental methods, and signal processing. In: *Handbook of Neuroimaging Data Analysis*, pp. 175–197.
- Nunez, P.L., Srinivasan, R., 2006. *Electric Fields of the Brain: The Neurophysics of EEG*, second ed. Oxford University Press, New York, <https://doi.org/10.1093/acprof:oso/9780195050387.001.0001>.
- Nunez, P.L., Srinivasan, R., Westdorp, A.F., Wijesinghe, R.S., Tucker, D.M., Silberstein, R.B., Cadusch, P.J., 1997. EEG coherency. I: Statistics, reference electrode, volume conduction, Laplacians, cortical imaging, and interpretation at multiple scales. *Electroencephalogr. Clin. Neurophysiol.* 9402881 103, 499–515.
- Nunez, P.L., Wingeier, B.M., Silberstein, R.B., 2001. Spatial-temporal structures of human alpha rhythms: theory, microcurrent sources, multiscale measurements, and global binding of local networks. *Hum. Brain Mapp.* 13, 125–164.
- Oohashi, T., Kawai, N., Honda, M., Nakamura, S., Morimoto, M., Nishina, E., Maekawa, T., 2002. Electroencephalographic measurement of possession trance in the field. *Clin. Neurophysiol.* 113, 435–445.
- Owen, J.P., Sekihara, K., Nagarajan, S.S., 2012. Non-parametric statistical thresholding for sparse magnetoencephalography source reconstructions. *Front. Neurosci.* 6, 186.
- Pascual-Marqui, R.D., 2002. Standardized low-resolution brain electromagnetic tomography (sLORETA): technical details. *Methods Find. Exp. Clin. Pharmacol.* 24 (Suppl. D), 5–12.
- Pascual-Marqui, R.D., Lehmann, D., Koenig, T., Kochi, K., Merlo, M.C., Hell, D., Koukkou, M., 1999. Low resolution brain electromagnetic tomography (LORETA) functional imaging in acute, neuroleptic-naïve, first-episode, productive schizophrenia. *Psychiatry Res.* 90, 169–179.
- Pascual-Marqui, R.D., Michel, C.M., Lehmann, D., 1994. Low resolution electromagnetic tomography: a new method for localizing electrical activity in the brain. *Int. J. Psychophysiol.* 18, 49–65.
- Pekala, R.J., Kumar, V.K., Maurer, R., Elliott-Carter, N., Moon, E., Mullen, K., 2010. Suggestibility, expectancy, trance state effects, and hypnotic depth: I. implications for understanding hypnotism. *Am. J. Clin. Hypn.* 52, 275–290. <https://doi.org/10.1080/00029157.2010.10401732>.
- Peres, J.F., Moreira-Almeida, A., Caixeta, L., Leao, F., Newberg, A., 2012. Neuroimaging during trance state: a contribution to the study of dissociation. *PLoS One* 7, e49360. <https://doi.org/10.1371/journal.pone.0049360>.
- Pernet, C.R., Chauveau, N., Gaspar, C., Rousselet, G.A., 2011. LIMO EEG: a Toolbox for Hierarchical Linear Modeling of ElectroEncephaloGraphic Data. *Comput. Intell. Neurosci.* 2011. <https://doi.org/10.1155/2011/831409>.
- Pernet, C.R., Latinus, M., Nichols, T.E., Rousselet, G.A., 2015. Cluster-based computational methods for mass univariate analyses of event-related brain potentials/fields: a simulation study. *J. Neurosci. Methods.* <https://doi.org/10.1016/j.jneumeth.2014.08.003>.

- Pernet, C.R., Appelhoff, S., Gorgolewski, K.J., Flandin, G., Phillips, C., Delorme, A., Oostenveld, R., 2019. EEG-BIDS, an extension to the brain imaging data structure for electroencephalography. *Sci. Data* 6, 103. <https://doi.org/10.1038/s41597-019-0104-8>.
- Picton, T.W., 2010. *Human Auditory Evoked Potentials*. San Diego, CA, USA, McNaughton and Gunn: Plural Publishing.
- Pieper, K., Spang, R.P., Prietz, P., Möller, S., Paajanen, E., Vaalgamaa, M., Voigt-Antons, J.-N., 2021. Working with environmental noise and noise-cancellation: a workload assessment with EEG and subjective measures. *Front. Neurosci.* 15.
- Plourde, G., 2006. Auditory evoked potentials. *Best Pract. Res. Clin. Anaesthesiol.* 20, 129–139. <https://doi.org/10.1016/j.bpa.2005.07.012>.
- Poerio, G.L., Totterdell, P., Miles, E., 2013. Mind-wandering and negative mood: does one thing really lead to another? *Consciousness Cogn.* 22, 1412–1421. <https://doi.org/10.1016/j.concog.2013.09.012>.
- Puiui, T., 2021. Inside the quietest room in the world where you can hear your heartbeat and blood flow. *ZME Sci.* URL <https://www.zmescience.com/other/feature-post/quietest-room-world/> (accessed 7.14.22).
- R Development Core Team, R, 2021. *R: A Language and Environment for Statistical Computing*.
- Rocha, A.C., Paraná, D., Freire, E.S., Lotufo Neto, F., Moreira-Almeida, A., 2014. Investigating the fit and accuracy of alleged mediumistic writing: a case study of chico Xavier's letters. *Explore (NY)* 10, 300–308. <https://doi.org/10.1016/j.explore.2014.06.002>.
- Rock, A.J., Krippner, S., 2012. States of consciousness or states of phenomenology? In: *Consciousness: States, Mechanisms and Disorders, Perspectives on Cognitive Psychology*. Nova Science Publishers, Hauppauge, NY, US, pp. 55–65.
- Rodriguez-Larios, J., Montes, B., de Oca, E.A., Alaerts, K., 2021. The EEG spectral properties of meditation and mind wandering differ between experienced meditators and novices. *NeuroImage* 245, 118669. <https://doi.org/10.1016/j.neuroimage.2021.118669>.
- Rose, N.L., Yang, H., Turner, S.D., Simpson, G.L., 2012. An assessment of the mechanisms for the transfer of lead and mercury from atmospherically contaminated organic soils to lake sediments with particular reference to Scotland, UK. *Geochim. Cosmochim. Acta* 82, 113–135.
- Sabourin, M.E., Cutcomb, S.D., Crawford, H.J., Pribram, K., 1990. EEG correlates of hypnotic susceptibility and hypnotic trance: spectral analysis and coherence. *Int. J. Psychophysiol.* 10, 125–142. [https://doi.org/10.1016/0167-8760\(90\)90027-B](https://doi.org/10.1016/0167-8760(90)90027-B).
- Sahdra, B.K., MacLean, K.A., Ferrer, E., Shaver, P.R., Rosenberg, E.L., Jacobs, T.L., Zanesco, A.P., King, B.G., Aichele, S.R., Bridwell, D.A., Mangun, G.R., Lavy, S., Wallace, B.A., Saron, C.D., 2011. Enhanced response inhibition during intensive meditation training predicts improvements in self-reported adaptive socioemotional functioning. *Emotion* 11, 299–312. <https://doi.org/10.1037/a0022764>.
- Savers, B.M., Beagley, H.A., Henshall, W.R., 1974. The mechanism of auditory evoked EEG responses. *Nature* 247, 481–483. <https://doi.org/10.1038/247481a0>.
- Scheeringa, R., Bastiaansen, M.C.M., Petersson, K.M., Oostenveld, R., Norris, D.G., Hagoort, P., 2008. Frontal theta EEG activity correlates negatively with the default mode network in resting state. *Int. J. Psychophysiol. Integr. EEG fMRI* 67, 242–251. <https://doi.org/10.1016/j.ijpsycho.2007.05.017>.
- Seligman, R., 2005. From affliction to affirmation: narrative transformation and the therapeutic of Candomblé mediumship. *Transcult. Psychiatr.* 42, 272–294.

- Seligman, R., Kirmayer, L.J., 2008. Dissociative experience and cultural neuroscience: narrative, metaphor and mechanism. *Cult. Med. Psychiatry* 32, 31–64.
- Shakil, S., Magnuson, M.E., Keilholz, S.D., Lee, C.-H., 2014. Cluster-based analysis for characterizing dynamic functional connectivity. *Conf. Proc. IEEE Eng. Med. Biol. Soc.* 2014, 982–985. <https://doi.org/10.1109/EMBC.2014.6943757>.
- Simpson, G.L., Singmann, H., 2022. Gratia: Graceful ‘ggplot’-Based Graphics and Other Functions for GAMs Fitted Using “mgcv”.
- Stephan, K.E., Harrison, L.M., Kiebel, S.J., David, O., Penny, W.D., Friston, K.J., 2007. Dynamic causal models of neural system dynamics: current state and future extensions. *J. Biosci.* 32, 129–144. <https://doi.org/10.1007/s12038-007-0012-5>.
- Stoll, F.M., Wilson, C.R.E., Faraut, M.C.M., Vezoli, J., Knoblauch, K., Procyk, E., 2016. The effects of cognitive control and time on frontal beta oscillations. *Cereb. Cortex* 26, 1715–1732. <https://doi.org/10.1093/cercor/bhv006>.
- Tahiri, M., Mottillo, S., Joseph, L., Pilote, L., Eisenberg, M.J., 2012. Alternative smoking cessation aids: a meta-analysis of randomized controlled trials. *Am. J. Med.* 125, 576–584. <https://doi.org/10.1016/j.amjmed.2011.09.028>.
- Tallon-Baudry, C., Bertrand, O., 1999. Oscillatory gamma activity in humans and its role in object representation. *Trends Cogn. Sci.* 3, 151–162.
- Thomason, T.C., 2010. The role of altered states of consciousness in Native American healing. *J. Rur. Commun. Psychol.* E13 (E13), 1–11.
- Thompson, T., Terhune, D.B., Oram, C., Sharangparni, J., Rouf, R., Solmi, M., Veronese, N., Stubbs, B., 2019. The effectiveness of hypnosis for pain relief: a systematic review and meta-analysis of 85 controlled experimental trials. *Neurosci. Biobehav. Rev.* 99, 298–310. <https://doi.org/10.1016/j.neubiorev.2019.02.013>.
- Travis, F., 2021. Negative and positive mind-wandering. *Advancing Innovation and Sustainable Outcomes in International Graduate Education*. IGI Global, pp. 177–192.
- Vago, D.R., Zeidan, F., 2016. The brain on silent: mind wandering, mindful awareness, and states of mental tranquility. *Ann. N. Y. Acad. Sci.* 1373, 96–113. <https://doi.org/10.1111/nyas.13171>.
- van Son, D., De Blasio, F.M., Fogarty, J.S., Angelidis, A., Barry, R.J., Putman, P., 2019. Frontal EEG theta/beta ratio during mind wandering episodes. *Biol. Psychol.* 140, 19–27. <https://doi.org/10.1016/j.biopsycho.2018.11.003>.
- Wagner, J., Wessel, J.R., Ghahremani, A., Aron, A.R., 2018. Establishing a right frontal beta signature for stopping action in scalp EEG: implications for testing inhibitory control in other task contexts. *J. Cogn. Neurosci.* 30, 107–118. https://doi.org/10.1162/jocn_a_01183.
- Wahbeh, H., Butzer, B., 2020. Characteristics of English-speaking trance channelers. *EXPLORE* 16, 304–309. <https://doi.org/10.1016/j.explore.2020.02.002>.
- Wahbeh, H., Carpenter, L., Radin, D., 2018. A mixed methods phenomenological and exploratory study of channeling. *J. Soc. Psych. Res.* 82, 129–148.
- Wahbeh, H., McDermott, K., Sagher, A., 2018a. Dissociative symptoms and anomalous information reception. *Act. Nerv. Super.* 1–11. <https://doi.org/10.1007/s41470-018-0023-6>.
- Wahbeh, H., Sagher, A., Back, W., Pundhir, P., Travis, F., 2018b. A systematic review of transcendent states across meditation and contemplative traditions. *EXPLORE* 14, 19–35. <https://doi.org/10.1016/j.explore.2017.07.007>.
- Wahbeh, H., Cannard, C., Okonsky, J., Delorme, A., 2019. A physiological examination of perceived incorporation during trance. *F1000Res.* 8, 67. <https://doi.org/10.12688/f1000research.17157.2>.

- Walters, S., 2020. Altering consciousness without drugs. In: *Psychology-1st Canadian Edition*. Thompson Rivers University, Canada.
- Wickham, H., 2016. Package ‘ggplot2’: Elegant Graphics for Data Analysis. Springer-Verlag, New York, https://doi.org/10.1007/978-3-319-24277-4_1.
- Wickham, H., François, R., Henry, L., Müller, K., RStudio, 2022a. dplyr: A Grammar of Data Manipulation.
- Wickham, H., Girlich, M., RStudio, 2022b. tidyr: Tidy Messy Data.
- Winkelman, M., 1986. Trance states: a theoretical model and cross-cultural analysis. *Ethos* 14, 174–203. <https://doi.org/10.1525/eth.1986.14.2.02a00040>.
- Winkelman, M., 2011. Shamanism and the alteration of consciousness. In: Cardeña, E., Winkelman, M. (Eds.), *Altering Consciousness Multidisciplinary Perspectives Volume 1 History, Culture and the Humanities*. Preager ABC-CLIO, Santa Barbara, CA, US, pp. 159–180.
- Wood, S.N., 2011. Fast stable restricted maximum likelihood and marginal likelihood estimation of semiparametric generalized linear models. *J. R. Stat. Soc. Ser. B* 73, 3–36.
- Xiao, N., Li, M., 2018. ggsci: Scientific Journal and Sci-Fi Themed Color Palettes for “ggplot2”.
- Zhou, X., Lei, X., 2018. Wandering minds with wandering brain networks. *Neurosci. Bull.* 34, 1017–1028. <https://doi.org/10.1007/s12264-018-0278-7>.